Investigating Depth-Based Target Expansion for Gaze Selection with Dwell Activation in Virtual Environments

Rumeysa Turkmen, Zahra Rasoulifar, Aunnoy K Mutasim, and Anil Ufuk Batmaz

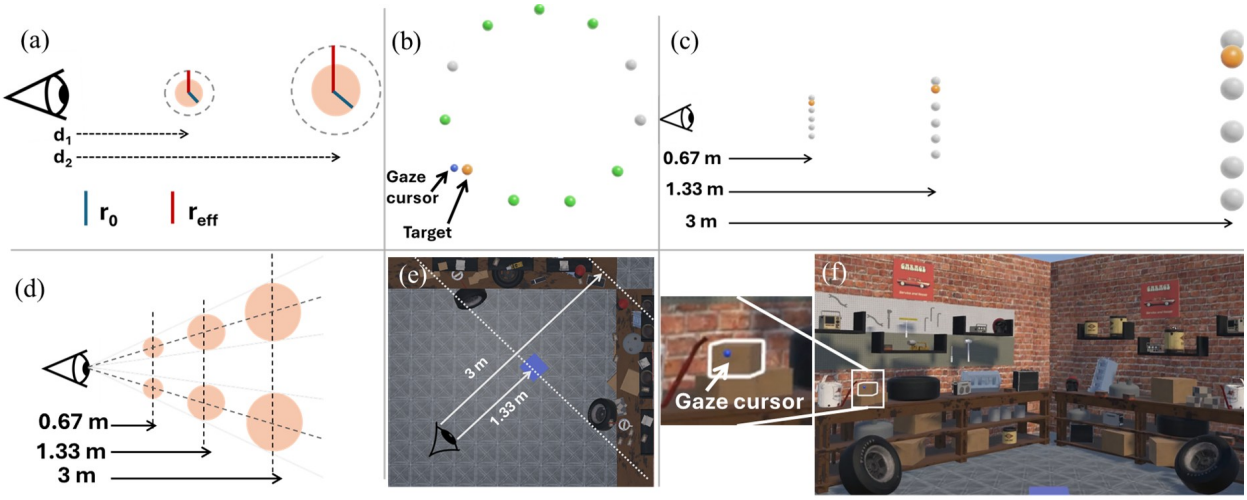


Figure 1: Overview of the proposed DEPTH-BASED expansion method and the experimental setups. (a) In DEPTH-BASED, farther targets receive larger effective activation radii than nearer targets. r_0 shows the base target radius, r_{eff} shows the expanded radius in motor-space without visual changes. Study 1 setup: (b) User view of the experiment. Blue sphere indicates the gaze cursor, orange is the current target, and green spheres are correctly selected targets. (c) Side isometric view of the ISO 9241:411 multidirectional layout for the same target size and target distance (in visual angles) across the three evaluated depths (0.67 m, 1.33 m, and 3 m). (d) We adjusted target sizes and target distances to be perceived in the same visual angle at each depth. Study 2 setup: (e) Top view of the garage environment showing the participant position, start plate (blue), and target distribution, and (f) participant view of the 3D scene with the blue gaze cursor and white outlined target.

Abstract—While dwell-based selection enables interaction with little physical effort, it is prone to (eye) gaze jitter. Target expansion methods mitigate this problem by increasing the effective selectable target area in 2D environments. Yet depth-related factors may limit their effectiveness in 3D virtual environments due to stereo-deficiencies. In this paper, we investigate how depth affects target expansion methods for dwell-based gaze selection, and propose DEPTH-BASED expansion, a method that adapts target expansion based on depth. We conducted two user studies to evaluate our proposed method with an ISO 9241:411 multidirectional selection study and a 3D selection study in a complex environment. Our results show that while the effectiveness of 2D target expansion methods decreases significantly with increasing depth, our proposed method consistently improved user performance and experience. Our findings provide design insights for effective dwell-based selection of expanding targets in 3D virtual environments.

Index Terms—Eye Gaze, 3D Selection, Virtual Reality, Target Expansion, Dwell.

1 INTRODUCTION

With the increasing integration of eye trackers into commercial head-mounted displays (HMDs), gaze has become a widely explored input modality for 3D interaction. Eye movements can reach velocities of up to 900° per second [4], enabling rapid and omnidirectional pointing through natural eye movements [12, 35, 56, 65]. Moreover, gaze allows users to interact with distant virtual or real-world objects without using their hands, enabling hands-free interaction in situations where users' hands may already be occupied with object manipulation or controllers [16]. In addition, gaze requires minimal muscular effort [62] and provides an immediate indicator of user attention [44, 70]. As a result, prior work has demonstrated that gaze-based pointing can be faster than hand, controller, or head-based interaction techniques in 3D environments [12, 35, 65].

While gaze is highly effective for target pointing, it does not provide a mechanism for explicit target selection, i.e., users naturally look at objects, not necessarily to select them. To address this limitation, researchers have combined gaze with additional input modalities to enable target selection, such as hands and controllers. Common approaches include pinch gestures [32, 52, 55], button clicks [50], speech input [7, 18], and dwell, where selection is triggered after maintaining fixation on the target for a predefined duration, i.e., the dwell time [17, 22, 40, 49, 50, 58]. Prior studies found that dwell is faster than clicks and pinch [29, 35]. Moreover, it does not require additional input for target selection, unlike, e.g., controllers, and allows hands-free interaction.

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However, human eyes are never perfectly stable [48]. Even during fixation, they produce microsaccades, tremor, and drift. These small involuntary movements, on top of eye-tracking accuracy and precision limitations [50], cause the gaze cursor to leave the target briefly, which can reset the dwell timer, and thus increase the selection time [49]. Moreover, dwell-based selection mechanisms introduce a trade-off between speed and accuracy during target selection. Short dwell times enable faster selections but increase unintended activations, whereas longer dwell times reduce false activations at the cost of slower interaction [38, 48, 62]. This performance trade-off makes dwell-based selection particularly challenging when targets are small, distant, and densely arranged, as even little gaze jitter can trigger wrong selections.

Target expansion is one of the strategies explored to mitigate dwell instability in selection tasks. Rather than reducing gaze jitter itself, expansion increases a target's effective selectable area, improving tolerance to gaze instabilities and reducing unintended activations. In 2D interfaces, such as gaze-based keyboards and menus, target expansion has been shown to significantly improve selection accuracy [43, 45, 73]. Yet, in 3D environments, depth perception can directly affect interaction performance [5, 8, 9]. One factor contributing to this effect is the vergence–accommodation conflict (VAC), a mismatch between vergence and accommodation that arises in most single-focal-plane VR displays and can degrade spatial perception and interaction performance [8, 9]. As targets are distributed across different depths, expansion methods may perform under different geometric and perceptual constraints compared to 2D interfaces. In other words, directly transferring the knowledge of 2D target expansion might not provide the same performance benefits and reliability at different depths.

In this paper, our goal is to understand how previously proposed 2D target expansion methods are affected by different target depths in Virtual Reality (VR). Further, here we propose a DEPTH-BASED expansion method that adapts the expansion behavior according to the depth of the target. Unlike prior expansion methods designed for 2D interfaces, our approach explicitly accounts for target depth. To understand the effect of depth on target expansion and evaluate DEPTH-BASED expansion, we conduct two user studies in VR: (1) an ISO 9241:411 multidirectional selection study and (2) a 3D selection study in a dense environment. Our contributions are:

- Showing target expansion methods designed for 2D interfaces cannot be directly applied to 3D VR environments, as target depth significantly affects dwell-based gaze selection performance.
- Proposing DEPTH-BASED expansion, a target expansion method that adapts expansion behavior according to target depth.
- Demonstrating DEPTH-BASED expansion maintains stable user performance across different target depths, consistently outperforming commonly used alternatives.
- Beyond VAC studies, to the best of our knowledge, presenting the first VR study that investigates depth effects on target expansion and interaction methods while controlling visual angle across depths.

2 RELATED WORK

2.1 Gaze-Based Selection

Researchers have explored several techniques for gaze-based selection, among which dwell is one of the most widely adopted gaze-only approaches [24, 32, 49, 50, 57, 77]. In dwell-based selection, a target is activated after the user maintains fixation on it for a predefined duration [22, 40, 58]. Dwell has been widely used in applications such as menu selection, scrolling, virtual keyboards, accessibility systems, and object manipulation [20, 34, 51, 57]. Although dwell has been found faster than hand- and controller-based raycasting in some settings, it suffers from the *Midas touch* problem, i.e., unintended target activations [30, 35, 65], and from gaze jitter during fixation, which can cause repeated exits from the target region before dwell completion [49].

Prior work has explored different ways to mitigate these problems. For instance, researchers adapted dwell time based on estimated user intent, such as Narkar et al.'s [53] GazeIntent, which adjusts dwell time using gaze-intent model outputs. Similarly, Isomoto et al. [28] proposed a two-step method in which the dwell duration must first exceed a static threshold, after which an intent prediction model confirms the selection. Although these methods reduce erroneous selections, they also increase interaction time, which may limit their usability in tasks with frequent selections. In addition, such methods rely on machine learning models and may not generalize well across tasks.

Magnification-based approaches have also been explored to support dwell-based selection, either by enlarging parts of the scene [1, 37, 42] or by facilitating the selection of small objects [2, 15, 63, 64]. In most of these systems, the magnified view is activated at or near the user's gaze position after a predefined dwell duration. For example, Lankford [36] proposed a two-step dwell-based method in which a zoom window is activated before the final selection. Ashmore et al. [2] introduced a fisheye lens to magnify local regions around the gaze point. Mutasim et al. [50] presented a magnified view of the gazed region, enlarged by a factor of four after a button press. Choi et al. [15] proposed BubbleGaze, which replaces the point cursor with a dynamically resizing circular area that automatically expands to include the nearest target, thereby enlarging the effective selectable region. All these works showed that magnification can support the accurate selection of very small targets. Despite its benefits, magnification-based approaches introduce an additional interaction step of magnifying the targets and shift users from direct to indirect target selection. This may increase interaction complexity, especially because scene changes during 3D selection can negatively affect user experience, immersion, and presence [71].

Prior work has also explored multi-modal gaze-based selection methods that combine gaze pointing with an additional confirmation input, such as hand gestures and voice [7]. One common example is Gaze+Pinch, in which users point with their gaze and confirm selection with a pinch gesture [52, 54, 55]. Similarly, gaze combined with mouse or controller clicks has also been explored as a confirmation mechanism [52, 77]. For example, Zhang et al. [77] proposed Gaze-Lock, in which the gaze cursor becomes locked to a target after remaining within its region for a predefined duration, thereby reducing eye jitter before final selection is confirmed with a mouse click. Although such techniques can reduce unintended activations, they require an explicit motor action in addition to gaze input. Thus, dwell remains the most widely used gaze-based selection technique, largely due to its potential in assistive technologies [49].

2.2 Target Expansion

Target expansion has been explored as a way to improve gaze-based target selection by increasing the effective selectable area of objects. In dwell-based interaction, this is particularly useful because selection reliability is highly sensitive to gaze jitter and fixation instability, especially when targets are small. Rather than reducing noise in the gaze signal itself, expansion improves tolerance to gaze imprecision during the selection process.

Prior work has investigated several forms of target expansion in gaze interfaces. Early studies explored static motor-space expansion, where the visual appearance of a target remains unchanged but its effective selectable region is enlarged to compensate for gaze instability [45]. Other work examined visible and dynamic expansion, showing that users can also benefit from targets that visually grow during interaction [43, 73]. In addition, methods such as the Grab-and-Hold Algorithm (GHA) were proposed to stabilize dwell selection by maintaining the selection state once gaze enters an expanded target area, thereby reducing the effect of involuntary eye movements [45]. Target expansion has also been applied in dense gaze-based interfaces such as virtual keyboards, where it helps preserve usability despite limited target size [19, 46].

Although these studies demonstrate that target expansion can improve gaze-based selection, they have been investigated primarily in 2D interfaces such as menus and toolbars [43, 72], and keyboards [19, 46]. In contrast, targets in 3D environments can appear at different depths, resulting in different visual angles and precision demands. This suggests that the effectiveness of a given expansion strategy may vary depending on target depth.

To the best of our knowledge, only one study in VR [60] has investi-

gated target expansion in raycasting-based interaction. However, they did not investigate how depth influences expansion behavior across targets placed at different distances. Therefore, investigating depth when designing target expansion is critical for improving dwell-based gaze selection in 3D environments.

2.3 Fitts' Law

Fitts' law is a foundational model for evaluating target-directed movements in Human Computer Interaction, relating movement time (MT) to the ratio between movement amplitude and target width, expressed as the Index of Difficulty (ID) [39]. The law captures the speed-accuracy trade-off in pointing tasks and has been validated across various motor effectors, including arm, wrist, and head movements [66,67]. It has also been used in ISO 9241:411 [27] to "evaluate methods for the design of physical input devices for interactive systems."

Gaze-based Fitts' law tasks have been investigated on 2D displays [14,28,59] as well as in HMDs [3,18,22,52]. While direct comparisons between gaze and other input modalities require explicit activation mechanisms such as dwell [59,61], prior work supports the use of Fitts' law modeling to characterize gaze-based selection performance [68,77].

In this paper, we use the angular version of Fitts' law [33] to keep the visual target sizes and target distances the same across depth conditions [8]. Using the angular model of Fitts' law [33], movement time is defined as:

$$MT = a + b \cdot \log_2\left(\frac{\alpha}{\omega^k} + 1\right) = a + b \cdot ID_A \tag{1}$$

where α is the angular distance between targets, ω is the angular target width, and k is a relative weight, typically set to 1 [8,33]. The logarithmic term defines the angular index of difficulty, denoted as ID_A, while a and b are empirically derived constants obtained through linear regression.

In addition to MT, we define throughput for angular movements, a combined measure of speed and accuracy based on effective task difficulty, following ISO 9241-411:2015 [27]:

$$THP = \frac{ID_{eA}}{MT} \tag{2}$$

where ID_{eA} (Equation 3) is the effective angular index of difficulty. It is computed from the effective angular movement amplitude α_e and the effective angular target width ω_e determined by the standard deviation between the selected position and the target center (SD_x), and characterizes the accuracy of the task performance [39], calculated as $\omega_e = 4.133 \times SD_x$.

$$ID_{eA} = \log_2\left(\frac{\alpha_e}{\omega_e^k} + 1\right) = \log_2\left(\frac{\alpha_e}{(4.133 \cdot SD_x)^k} + 1\right) \tag{3}$$

3 Motivation

In 3D virtual environments, targets can be positioned at different depths, introducing perceptual and interaction-related factors that may affect gaze-based selection performance [5]. Although target expansion improves tolerance to gaze jitter, existing methods were developed for 2D interfaces and assume depth-independent performance. This assumption does not hold in HMDs, as stereo display deficiencies reduce depth perception and motor precision as viewing distance increases [8]. As a result, expansion strategies designed for 2D interfaces cannot be directly applied to targets distributed across 3D space.

To address this limitation, we investigate how target depth affects the performance of existing target expansion methods in dwell-based gaze selection, and propose Depth-Based expansion, a depth-aware method that adapts the expansion size according to the target depth. More specifically, here we investigate the following research questions:

RQ1: How does target depth affect the effectiveness of target expansion methods in dwell-based gaze selection in VR?

RQ2: Does adapting the expansion size according to target depth improve user performance and experience in dwell-based gaze selection?

To answer these questions, we conduct two user studies. In Study 1, we use an ISO 9241:411 multidirectional selection task to examine how depth influences the performance and user experience of existing expansion methods and our proposed method. In Study 2, we evaluate the Depth-Based expansion methods in a more practical 3D environment, considering both user performance and experience.

4 Target Expansion Techniques

4.1 Existing Target Expansion Techniques

Prior work investigated motor-space target expansion methods in which the effective activation region of a target is enlarged without visually altering its geometry [43], i.e., the expansion is not visible to the user. We chose motor-space expansion as we do not want to distort the user perception, as prior work on 3D selection found that making changes in the scene during target selection negatively affects the user performance [71]. Based on these findings, we implemented two expansion methods: Instant and Linear, which were explored in prior work [43,45] in 2D desktops. Previous studies also explored different expansion factors and reported that a factor of 3 was effective for improving selection performance [43,45,60]. Accordingly, we adopted an expansion factor of 3 in angular measures for both methods and applied the same factor across all depth conditions. Here, the expansion factor refers to the ratio between the expanded activation radius and the original target radius, meaning that the effective activation region can grow up to three times the original size.

In Instant, the target's effective activation radius expands immediately without any delay to its predefined maximum expansion factor once the gaze cursor enters the base target region. The expansion remains active until the gaze exits the expanded region.

In Linear, the activation radius increases linearly as a function of dwell time. Once the gaze cursor enters the base target region, the expansion factor interpolates linearly from 1 (original size) to the maximum expansion factor (3x) over a dwell duration. Similar to Instant, the expansion remains active until the gaze leaves the expanded region.

4.2 Depth-Based Target Expansion

We propose Depth-Based expansion, which adapts according to the target depth. Unlike prior methods that apply a fixed expansion factor irrespective of viewing distance, our method adaptively modulates the expansion factor as a function of target depth.

Let r_0 denote the base collider radius of a target in world space. We define the effective activation radius $r_{eff}(t,d)$ as:

$$r_{eff}(t,d) = r_0\left[1 + (E_F - 1)\, f(t)\, g(d)\right] \tag{4}$$

where E_F is the base maximum expansion factor, $f(t)$ is the temporal growth term, and $g(d)$ is a depth-dependent multiplier. The temporal component $f(t)$ follows a normalized growth curve over the dwell interval. The depth component $g(d)$ compensates for increasing precision demands at larger viewing distances. We model this depth multiplier as a non-linear function, motivated by prior work showing that the effect of visual depth on 3D selection performance in VR is itself non-linear [5]. Accordingly, we do not assume that the increase in selection difficulty with depth is uniform across viewing distances. To make the expansion independent of absolute scene scale, depth is normalized between the minimum and maximum interaction distances:

$$d_{norm} = \frac{d - d_{close}}{d_{far} - d_{close}} \tag{5}$$

where d_{close} and d_{far} represent the minimum and maximum interaction distances in the scene. The depth multiplier is then defined as:

$$g(d) = 1 + \beta\left(e^{d_{norm}} - 1\right) \tag{6}$$

where β controls the strength of depth-based expansion. Thus, targets positioned farther from the user receive proportionally larger motor-space expansion.

Based on our pilot testing, we set $\beta = 0.5$, which provided a noticeable depth-dependent increase in expansion without causing overly aggressive growth for distant targets. This parameter can be further adjusted based on task requirements and interaction ranges, allowing the method to generalize across different application scenarios. The minimum and maximum interaction distances were defined separately for each study based on the task setup. In Study 1, d_{close} and d_{far} were set to 0.67 m and 3 m, respectively, whereas in Study 2 they were set to 1.33 m and 3 m. These depths were chosen to introduce depth variation around the Meta Quest Pro focal distance (1.33 m). We set E_F to 3, as prior work has shown this expansion factor to be effective across different target expansion settings [43, 45, 60]. To prevent unintended overlap between neighboring targets, the effective radius is capped by a nearest-neighbor constraint:

$$r_{eff} \leq \alpha \cdot \frac{d_{nn}}{2} \tag{7}$$

where d_{nn} is the center-to-center distance to the nearest neighboring target and $\alpha \in (0,1)$ is a safety coefficient. This coefficient keeps the expanded activation radius below the maximum non-overlapping radius, leaving a small margin between adjacent targets even under expansion. In our implementation, we set $\alpha = 0.9$. In both studies, we ensured that targets did not overlap even under maximum expansion; however, this constraint provides an additional safety margin and improves the generalizability of the technique to denser settings.

We note that α acts as a density-aware safety constraint rather than a performance-tuning parameter: it does not control the rate of depth-based expansion, but only bounds the maximum allowable expansion radius to prevent overlap with neighboring targets (Equation 7). With $\alpha = 0.9$, expansion is permitted up to 90% of the distance to the nearest neighbor, leaving a 10% safety margin, regardless of α's specific value beyond this collision constraint. In contrast, β controls the strength of depth-compensation: increasing β produces stronger expansion for farther targets, while decreasing β makes the method behave closer to a fixed expansion factor. The same parameter values ($\beta = 0.5$, $\alpha = 0.9$) generalized across both studies despite differing scene layouts (a controlled ISO 9241:411 task and a denser garage scene), suggesting reasonable robustness across settings.

In pilot testing, we also evaluated quadratic depth-scaling formulation. Quadratic scaling over-compensated at near depths, motivating our choice of an exponential formulation that provides a smoother, more gradual increase near the user while still compensating for distant targets.

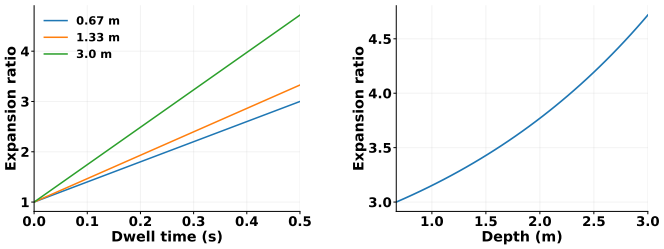
5 User Studies

We conducted two user studies to evaluate the effect of target depth on dwell-based gaze selection and to assess the performance of our proposed Depth-Based expansion method. Study 1 is a controlled ISO 9241:411 multidirectional selection study that isolates the effect of depth across three viewing distances (0.67 m, 1.33 m, and 3 m). Study 2 provides ecological validation by evaluating Depth-Based against commonly used gaze-based selection techniques in a visually complex virtual garage environment.

Both user studies reported in this paper were conducted under the same ethics approval. The study protocols were reviewed and approved by Concordia University's Human Research Ethics Committee (Certification Number: 30021253). All participants provided informed consent prior to participation.

6 Study 1

We conducted an ISO 9241:411 multidirectional selection study to understand the effect of depth on different expansion methods for dwell-based selection. We compared our proposed expansion method Depth-Based with Linear, Instant, and NoExp (where there is no expansion during dwell selection). For Depth-Based, as shown in Figure 2(a), the expansion ratio increases progressively over dwell time, with farther targets receiving larger effective activation regions. Figure 2(b) further illustrates that the expansion ratio increases non-linearly as a function of target depth.



(a) (b)

Figure 2: Depth-Based expansion behavior during Study 1: (a) Expansion over the dwell time and (b) Expansion ratio for each depth.

6.1 Participants

24 participants (11 female and 13 male) with ages ranging between 21 and 43 (M = 25.9, SD = 4.35) from the local university participated in Study 1. All participants reported normal or corrected-to-normal vision. When asked about how many times they had experienced VR previously, four participants reported never, ten reported 1-3 times, and ten participants reported more than 5 times.

6.2 Apparatus

We used a 13th Gen Intel(R) Core(TM) i9-13900KF at 5.8 GHz, 32 GB RAM desktop PC with an NVIDIA GeForce RTX 4070 graphics card. We used a Meta Quest Pro HMD (field of view: 106° horizontal × 96° vertical; display resolution: 1800 × 1920 pixels per eye; refresh rate: 90 Hz), and Unity 2021.3.24f1 to design and implement the virtual environment.

6.3 Experimental Design and Evaluation Metrics

We conducted a within-subjects user study to investigate how different target expansion strategies perform under varying depth conditions in dwell-based selection. We compared four **Expansion Methods** (4_{EM} = Depth-Based, Linear, Instant and NoExp) and three **Depths** ($3_D = 0.67$ m, 1.33 m, and 3 m) yielding a ($4_{EM} \times 3_D =$) 12 condition (12_{cond}) design. We counterbalanced them across participants using a Latin Square. In total, we used 6 unique ID_As, based on three **angular target sizes** ($3_{ATS} = 0.85°$, 1.69°, and 2.96°) and two **angular target distances** ($2_{ATD} = 16.80°$ and 24.99°). Using Kopper et al.'s [33] angular size equation, we adjusted the physical target sizes according to the viewing distance so that all targets had the same visual angle and appeared identical in size across depths, as shown in Figure 1(d). Each participant performed $12_{cond} \times 6_{ID_A} \times 11$ repetitions = 792 trials. The dwell duration was set to 500 ms for all methods. This value was chosen as a mid-range dwell duration, since prior work has not established a single optimal threshold and commonly reports dwell times between 300 and 1000 ms [52].

Corresponding to the ISO9241:411 multidirectional selection task, participants were presented with a circle of 11 gray spheres and were instructed to select the target sphere highlighted in orange (see Figure 1(b). Selection was triggered when the gaze cursor remained within the target for the required dwell time. A successful selection turned the sphere green, whereas an unsuccessful one turned it red. After each trial, the next target in the circular sequence was highlighted. The gaze cursor size was set to 0.5°, which is smaller than the smallest target, and its visual angle was adjusted across depths to maintain perceptual consistency, following prior recommendations [74]. Pilot testing showed that participants struggled to complete selection in the NoExp technique for small targets. Since the interaction did not include any manual confirmation mechanism (e.g., button press), some trials could last excessively long when participants failed to stabilize their gaze on the target. We therefore introduced a 15-second timeout: trials that were not completed within this period were automatically terminated, recorded as errors, and the system proceeded to the next target. To avoid skewing the analysis with these extreme outliers, Movement Time (MT)

was computed after excluding such timeout trials. In contrast, Total Time (TT) includes the full duration of all trials, including those that reached the timeout threshold. Aside from MT and TT, we analyzed error rate, angular SD_x, and angular throughput (bits/s), as in prior work [5, 8, 10].

6.4 Procedure

Participants were given a brief overview of the study and asked to read and sign a consent form before proceeding. Before starting the experiment, participants were asked to adjust the headset's interpupillary distance (IPD) for optimal visual comfort, and for each expansion method condition, the eye tracker was calibrated. Participants were not required to move their heads, as also applied in prior studies [6, 10, 11]. After completing each condition, participants filled out three subjective questionnaires: the NASA Task Load Index (NASA-TLX) [23], the System Usability Scale (SUS) [13], and the short version of the User Experience Questionnaire (UEQ-S) [25]. After completing all experimental conditions, participants filled out a post-experiment questionnaire to provide feedback about their experience. The experiment lasted about one hour per participant.

6.5 Results

The data was pre-processed and plotted through JMP software. Data analysis was done using two-way repeated measures (RM) ANOVA using SPSS 29. The data was considered normally distributed if the Skewness (S) and Kurtosis (K) values were within ± 1 [21, 41]. Otherwise, we used log-transform before ANOVA. If the data were not normally distributed after the log transform, we used Aligned Rank Transformation (ART) [75] on the original data before ANOVA and applied ART-C [75] for pairwise comparisons. For non-parametric data, we analyzed it using Friedman's test and Wilcoxon signed-rank tests with Bonferroni correction for the post hoc analyses. For *brevity*, we only focus on significant results (see Table 1), specifically the effect of *Method* and the interaction between *Method* $\times$ *Depth*. Total Time results are aligned with the Movement Time results; therefore, Total Time results are reported in the supplementary materials. All pairwise post-hoc comparisons (exact *p*-values) for the results reported below are provided in the supplementary materials as well. Here we summarize the overall patterns supported by the figures.

Table 1: RM ANOVA Results with significant results shown in bold.

Parameters	Method (M)	Depth (D)	M $\times$ D
Movement Time (s)	**F(3,69)=24.733, p<0.001, η^2=0.518**	**F(2,46)=93.436, p<0.001, η^2=0.802**	**F(6,138)=19.47, p<0.001, η^2=0.05**
Error Rate	**F(3,69)=7.492, p<0.001, η^2=0.246**	**F(2,46)=15.08, p<0.001, η^2=0.396**	**F(6,138)=35.67, p<0.001, η^2=0.608**
Throughput (bits/s)	**F(3,69)=11.47, p<0.001, η^2=0.338**	**F(2,46)=50.214, p<0.001, η^2=0.686**	**F(6,138)=2.226, p=0.04, η^2=0.088**
SD_x	**F(3,69)=15.356, p<0.001, η^2=0.4**	F(2,46)=3.282, p=0.05, η^2=0.125	**F(6,138)=4.413, p=0.02, η^2=0.16**

6.5.1 Performance Results

Movement Time In terms of the effect of *Method*, participants were faster with DEPTH-BASED than with all other methods, and both LINEAR and INSTANT were faster than NOEXP (Figure 3(a)). The *Method* $\times$ *Depth* interaction showed that all methods slowed with increasing depth, but DEPTH-BASED was the least affected, showing a significant slowdown only between 0.67m and 3m, whereas the other methods slowed significantly at every depth step (Figure 3(b)). Consequently, DEPTH-BASED's advantage over the other methods widened at 1.33m and 3m compared to 0.67m (Figure 3(c)).

Error Rate The effect of *Method* showed that participants made significantly fewer errors with DEPTH-BASED than with all other methods, and error rates were also lower with LINEAR and INSTANT than with NOEXP (Figure 3(d)). The *Method* $\times$ *Depth* interaction showed that error rates increased with depth for NOEXP, INSTANT, and LINEAR, but this depth-related increase was not observed for DEPTH-BASED (Figure 3(e)). As a result, DEPTH-BASED maintained a significantly lower error rate than all other methods at 3m, while NOEXP produced the highest error rate at 1.33m and 3m (Figure 3(f)).

Throughput Participants achieved higher throughput with DEPTH-BASED than with all other methods (Figure 3(g)). The *Method* $\times$ *Depth* interaction showed that throughput decreased with depth for all methods, but this decrease was smaller for DEPTH-BASED, which only differed significantly between 0.67m and 3m (Figure 3(h)). DEPTH-BASED consistently outperformed all other methods at every depth level, and this advantage was present even at 0.67m (Figure 3(i)).

SD_x The standard deviation along the task axis was significantly lower with DEPTH-BASED than with all other methods (Figure 3(j)). The *Method* $\times$ *Depth* interaction showed that SD_x increased with depth for NOEXP, INSTANT, and LINEAR, but remained stable across depths for DEPTH-BASED (Figure 3(k)). Accordingly, DEPTH-BASED resulted in significantly lower SD_x than NOEXP at every depth, and outperformed all methods at 3m (Figure 3(l)).

Detailed Gaze Analysis We also analyzed detailed gaze behavior metrics reported in prior gaze interaction research [49] (see supplementary materials for full results). Overall, these measures showed a consistent pattern supporting the main findings. In particular, DEPTH-BASED resulted in fewer drop-offs and shorter drop-off durations, indicating that participants were less likely to leave the target region before completing the dwell. Similarly, activation time and on-target time were lower with DEPTH-BASED, suggesting more stable gaze behavior once the target was reached.

6.5.2 Questionnaire Results

Usability We found a significant main effect of *Method* on *SUS Score* ($\chi^2(3) = 64.769, p < .001$). DEPTH-BASED resulted in significantly higher SUS scores than all other methods, and both INSTANT and LINEAR also scored significantly higher than NOEXP (Figure 4(a)).

Task Load NASA-TLX results showed significant effects of *Method* on Mental Demand (MD) ($\chi^2(3) = 56.874, p < .001$), Physical Demand (PD)($\chi^2(3) = 45.695, p < .001$), Temporal Demand (TD) ($\chi^2(3) = 41.335, p < .001$), Effort ($\chi^2(3) = 58.739, p < .001$), Frustration ($\chi^2(3) = 51.377, p < .001$), Performance ($\chi^2(3) = 62.241, p < .001$), and Overall Workload ($\chi^2(3) = 65.184, p < .001$). Across every subscale, NOEXP was rated as most demanding, typically followed by INSTANT and LINEAR, with DEPTH-BASED consistently rated as the least demanding method. DEPTH-BASED was rated significantly lower (i.e., better) than INSTANT and LINEAR on Effort, and lower than INSTANT on Frustration; on the remaining subscales, DEPTH-BASED was significantly better than all other methods (Figure 4(e)). Full pairwise comparisons for each subscale are reported in the supplementary materials.

User Experience We found significant main effects of *Method* on *Pragmatic Quality* ($\chi^2(3) = 59.850, p < .001$), *Hedonic Quality* ($\chi^2(3) = 62.600, p < .001$), and *Overall UEQ-S score* ($\chi^2(3) = 64.550, p < .001$). DEPTH-BASED resulted in significantly higher (i.e., better) scores than all other methods on every measure, while both INSTANT and LINEAR also scored significantly higher than NOEXP (Figures 4(b), 4(c), and 4(d)).

7 STUDY 2

Study 2 aims to provide an ecological validation by examining whether the performance benefits of our proposed DEPTH-BASED method generalize to a realistic and visually complex virtual environment. In Study 2, we compare DEPTH-BASED expansion method with commonly used gaze-based selection techniques using the same apparatus in Study 1.

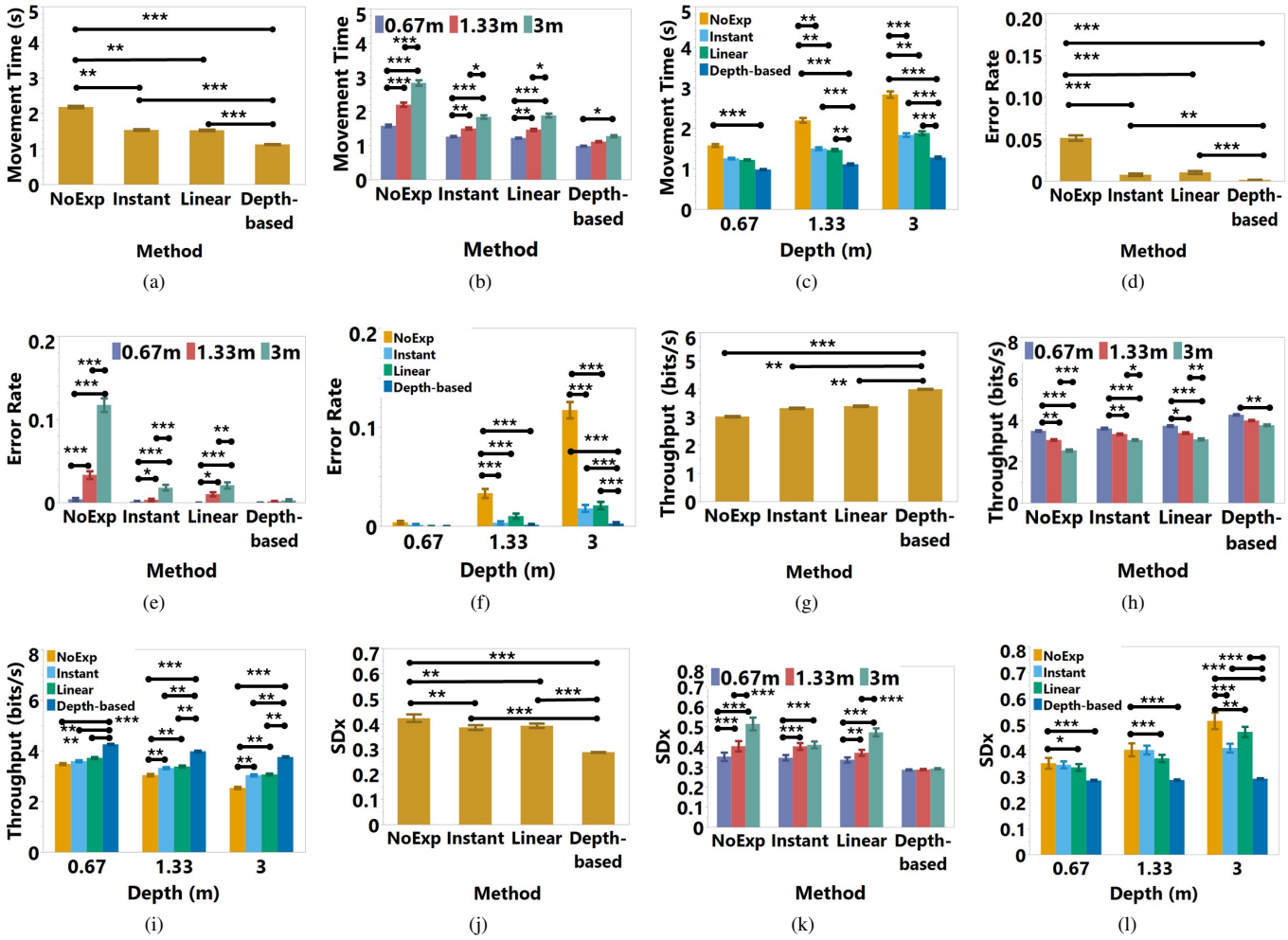


Figure 3: Study 1 performance results. For each metric, figures shows the main effect of Method (a, d, g, j), comparisons across depth levels within each method (b, e, h, k), and comparisons across methods within each depth level (c, f, i, l), respectively. Error bars indicate standard error. Significance levels are denoted as *** ($p < 0.001$), ** ($p < 0.01$), and * ($p < 0.05$).

7.1 Participants

We recruited 18 new participants (9 female, 9 male), ages ranging between 22 and 29 (M = 25.39, SD = 2.03). All participants reported normal or corrected-to-normal vision.

7.2 Environmental Settings

We designed a virtual garage environment (Figure 1(f)) in which selectable objects were distributed between 1.33m and 3m. In total, 75 objects were placed, and all objects in the scene, such as walls, floor, tools, desks, etc., were selectable. Among these, 30 objects were predefined as target candidates. To create a balanced spatial distribution, 15 target objects were placed on one side of the room, and their counterparts were placed on the opposite side, either as the same object or as another object of a similar size. For example, the red "Garage" plate appeared on both sides of the room. The scene was designed so that all targets were visible within (Angular target distance=24.99/2) 12.5 degrees from the center (about half of near peripheral) [26, 31], i.e., participants were not required to move their head to search for targets, as also applied in [5, 9, 11]. A blue plate was placed on the floor 1.33m from the participant. Participants were required to first fixate on this blue start plate, after which one target was highlighted. After completing the selection, they had to return their gaze to the start plate to trigger the next target. This procedure allowed us to standardize trial initiation across selections, thereby ensuring accurate selection time calculations.

The nearest-neighbor distance d_{nn} among target objects ranged from approximately 5–19 cm. With $\alpha = 0.9$, this corresponds to a maximum allowable expanded radius of approximately 2.3–8.6 cm before the overlap-prevention constraint Equation 7 is applied.

7.3 Experimental Design and Evaluation Metrics

We compared our method against four different gaze-based selection techniques *Gaze + Pinch* [52,55], *GazeLock* [77], *BubbleGaze* [15], and *Dwell (NoExp)*. These techniques were selected to represent different design strategies for gaze-based selection: GAZE+PINCH as a common multimodal confirmation technique, GAZELOCK as a stabilization-based approach for reducing gaze jitter, BUBBLEGAZE as a dynamic target-facilitation technique that enlarges the effective selectable region around the gaze point, NOEXP as the baseline dwell condition. Although NOEXP did not perform well in Study 1, we included it in Study 2 because the latter used a different task and a more practical 3D environment.

GAZE+PINCH requires users to aim at targets using gaze and confirm the selection using a pinch gesture (see Figure 5(a)). This technique represents a common multimodal interaction approach in XR systems where gaze is used for pointing and a separate manual input is used for confirmation [32, 52, 54, 55].

GAZELOCK (see Figure 5(b)) extends the conventional gaze cursor with a locking mechanism to reduce the effect of eye jitter. We followed the implementation of previous work [76], which was designed for 2D interfaces. When the gaze cursor remains within a target for longer than a predefined locking time (T_{lock}), the cursor becomes locked to the

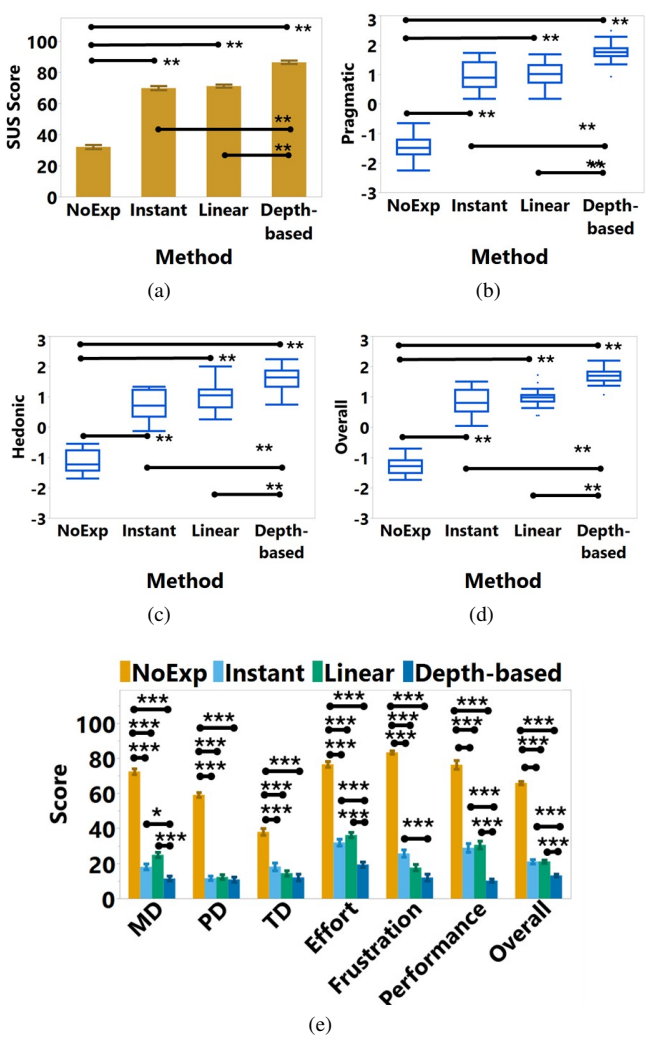


Figure 4: Study 1 Questionnaire Results: (a) SUS. UEQ metrics: (b) Pragmatic, (c) Hedonic, (d) Overall. (e) NASA-TLX scores. Error bars indicate standard error. Significance levels are denoted as *** ($p < 0.001$), ** ($p < 0.01$), and * ($p < 0.05$).

target center. Once locked, the cursor stays at the target center as long as the gaze remains within the target region. The cursor is not immediately released when the gaze leaves the target. Instead, it is unlocked only if the gaze remains outside the target longer than a predefined unlocking time (T_{unlock}). If the gaze returns to the target before this threshold, the cursor remains locked. Following the original technique, we set both T_{lock} and T_{unlock} to 100 ms. In our implementation, selection was confirmed using a pinch gesture after the cursor had been locked.

BUBBLEGAZE technique [15] adapts the size of a circular selection region around the gaze cursor based on the spatial arrangement of nearby objects (see Figure 5(c)). The bubble dynamically expands/contracts to include the nearest selectable object, and confirmation is done with dwell. We implemented BubbleGazeCursor rather than BubbleGazeLens, because BubbleGazeLens visually extends the target itself. In contrast, our study focuses on techniques that do not alter the target's visual shape.

We included our proposed method DEPTH-BASED and no expansion technique (here we refer to as NOEXP). We evaluated DEPTH-BASED with two dwell durations, 400ms (DEPTH-BASED$_{400}$) and 500ms (DEPTH-BASED$_{500}$), to examine whether its performance remains robust under a shorter dwell threshold. We selected 500ms to remain consistent with Study 1, and included 400ms because shorter dwell times are generally associated with faster interaction but also a higher risk of unintended activations [45]. This comparison allowed

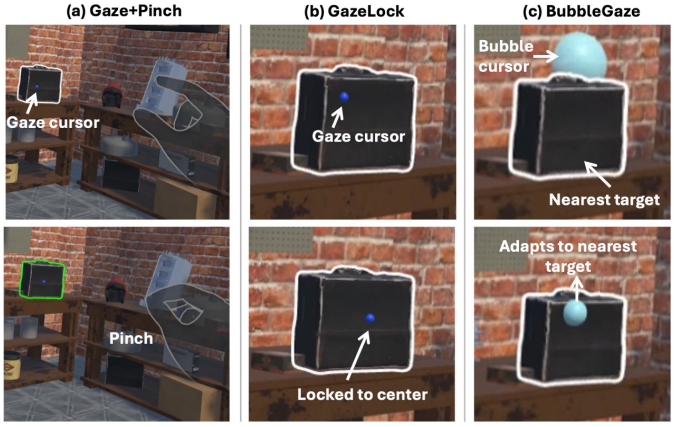


Figure 5: Selection techniques compared in Study 2. (a) GAZE+PINCH: the user aims with the gaze cursor (top) and confirms selection with a pinch gesture, turning the target outline green (bottom). (b) GAZELOCK: once the gaze cursor remains within the target long enough (top), it locks to the target center (bottom) and stays locked until the gaze exits the target region for longer than the unlocking threshold. (c) BUBBLEGAZE: a bubble cursor is displayed at the nearest selectable object to the gaze point (top); the bubble dynamically adapts as the gaze cursor moves toward a different nearest target (bottom).

us to assess whether the benefits of DEPTH-BASED persist when the dwell duration is reduced. In total, each participant performed the task with six **Techniques** (6_T = DEPTH-BASED$_{400}$, DEPTH-BASED$_{500}$, GAZE+PINCH, GAZELOCK, BUBBLEGAZE, and NOEXP) and they performed 30 target selections with each technique.

We recorded selection time (s) and error rate. Selection time started when participants fixated on the start point and ended when the target was successfully selected. The error rate was calculated based on incorrect selections.

7.4 Procedure

Participants were briefed about the study, signed the consent form, and filled out the questionnaire. Then, they were introduced to the six selection techniques. Before each technique, they completed a short practice session to become familiar with the interaction method. The order of the techniques was counterbalanced across participants. For each technique, participants performed 30 target selections. We randomized the target selection order for each selection technique to prevent participants from memorizing the sequence. After completing one technique, participants filled NASA-TLX and SUS questionnaires, and they could take a short break before continuing to the next condition. This procedure was repeated until all six techniques were completed. At the end of the study, participants filled post-experiment questionnaire. Overall, the study lasted about 40 minutes per participant.

7.5 Results

We followed the same analysis procedure as in Study 1. Since target sizes cannot be controlled in terms of visual angle in Study 2, we focus the main effects of Technique. Therefore, interaction between Technique $\times$ Depth analyses are provided in the supplementary materials.

7.5.1 Performance Results

We found significant effect of *Technique* on Total Time ($F_{5,80} = 13.593, p < .001, \eta^2 = .459$). NOEXP resulted in significantly slower selection compared to all techniques (Figure 6(a)). Moreover, Error Rate was found significantly affected by *Technique* ($F_{5,80} = 7.445, p < .001, \eta^2 = .318$). Both DEPTH-BASED techniques resulted in significantly fewer errors compared to GAZELOCK, GAZE+PINCH, BUBBLEGAZE, and NOEXP. Moreover, GAZELOCK, GAZE+PINCH, and NOEXP resulted in fewer errors compared to BUBBLEGAZE (Figure 6(b)).

7.5.2 Questionnaire Results

Usability As given in Table 2, SUS results showed that ($\chi^2(3) = 90.123, p < .001$) participants found BUBBLEGAZE less usable than DEPTH-BASED$_{400}$ ($p < .001$), DEPTH-BASED$_{500}$ ($p < .001$), GAZELOCK ($p < .001$), GAZE+PINCH ($p < .001$) and NOEXP ($p < .001$) (Figure 6(c)).

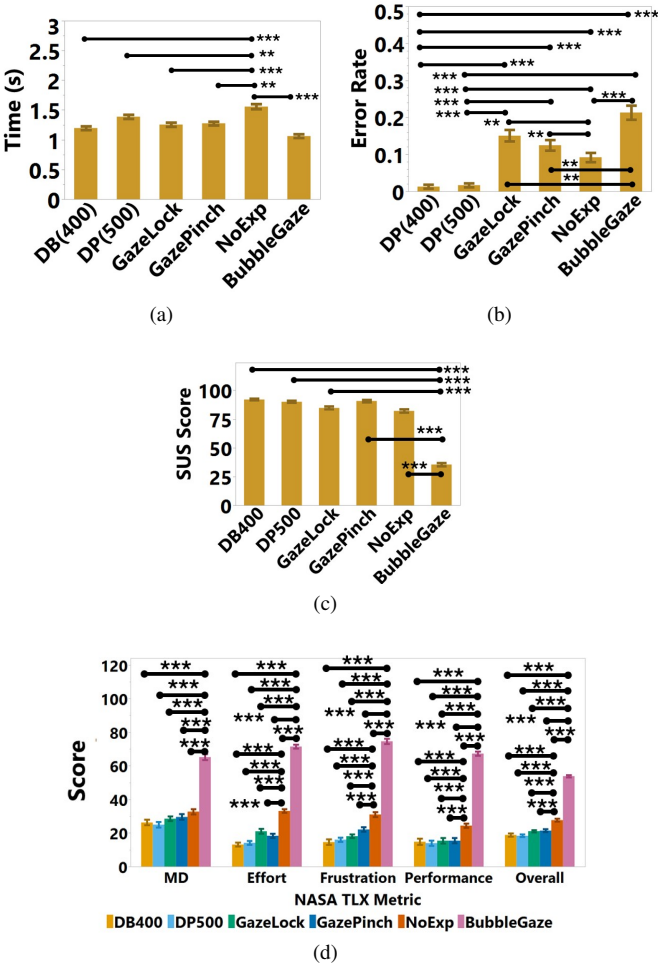


Figure 6: Study 2 Results: (a) Time, (b) Error Rate, (c) SUS, and (d) NASA-TLX. Error bars indicate standard error. Significance levels are denoted as *** ($p < 0.001$), ** ($p < 0.01$), and * ($p < 0.05$). DB400 and DB500 refer to DEPTH-BASED$_{400}$ and DEPTH-BASED$_{500}$, respectively.

Table 2: SUS Results and Grades.

Technique	SUS Score (M)	SUS Grade
DEPTH-BASED$_{400}$	91.98	A (Excellent)
DEPTH-BASED$_{500}$	90.04	A (Excellent)
GAZELOCK	84.69	B (Good)
GAZE+PINCH	90.73	A (Excellent)
NOEXP	82.08	B (Good)
BUBBLEGAZE	35.52	D (Poor)

Task Load We found significant main effects of *Technique* on Mental Demand ($\chi^2(5) = 67.159, p < .001$), Effort ($\chi^2(5) = 109.439, p < .001$), Frustration ($\chi^2(5) = 67.761, p < .001$), Performance ($\chi^2(5) = 88.425, p < .001$), and Overall Workload ($\chi^2(5) = 95.984, p < .001$). For Mental Demand, BUBBLEGAZE was rated significantly higher than all other techniques. For the remaining subscales (Effort, Frustration, Performance, and Overall Workload), both BUBBLEGAZE and NOEXP were rated significantly higher than DEPTH-BASED$_{400}$, DEPTH-BASED$_{500}$, GAZELOCK, and GAZE+PINCH, and BUBBLEGAZE was additionally rated higher than NOEXP on every subscale except Mental Demand (Figure 6(d)). Full pairwise comparisons are reported in the supplementary materials.

User Feedback We asked participants to provide their feedback about each selection technique. P6 commented *"[Depth-Based] was easier to select and I felt like it is the most straightforward technique."* For GAZELOCK, P9 commented *"... sometimes I felt like it locked unintentionally and made me select the wrong object."* P10 mentioned *"[Gaze+Pinch] was fast and easy, but having to pinch every time made it feel a bit more effortful than dwell interaction."* P2 stated *"[NoExp] was harder to keep the cursor on the target for smaller and farther objects."* About BUBBLEGAZE, P4 mentioned *"I found it confusing and mentally demanding. The changing bubble made it hard to predict what would actually be selected."*

8 DISCUSSION

This paper investigates how target depth affects the performance of dwell-based target expansion methods in 3D virtual environments. We also proposed and evaluated the DEPTH-BASED target expansion method that adjusts the selectable area based on the target depth.

The results of Study 1 show that target depth significantly affects the effectiveness of standard expansion methods for dwell activation. For the NOEXP, INSTANT, and LINEAR conditions, we observed that even when visual angles were kept constant across all depths, movement time, error rate, and SD_x increased while throughput decreased significantly. This suggests that the difficulty of maintaining a stable dwell fixation is not merely a function of visual size, but is affected by the perceptual and geometric constraints of 3D space, answering **RQ1 (How does target depth affect the effectiveness of target expansion methods in dwell-based gaze selection in VR?)**. Specifically, 2D INSTANT and LINEAR [43,45] methods showed a significant decrease in performance at farther depths. These results indicate that a fixed expansion strategy does not provide the same benefit at different viewing distances. In this context, the problem is not that prior 2D expansion methods are ineffective, as earlier work showed that target expansion can improve gaze-based selection in 2D interfaces [45,73]. However, these studies did not examine how such strategies perform when target depth varies. As a result, their findings cannot be generalized to scenarios where targets are distributed in depth and interaction is affected by stereo display deficiencies, such as the vergence–accommodation conflict (VAC) [8,11].

For DEPTH-BASED, the participants were slower and had a lower throughput at 3m compared to 0.67m, which can also be explained by the detrimental effects of VAC on user performance [8,9], i.e., the mismatch between the vergence distance of the virtual target and the fixed accommodation distance of the display. Yet, we did not observe any significant differences for error rate and SD_x across depth conditions. The error rate results can be explained by the enlarged motor-space activation region, which increases the tolerance to small gaze jitters and makes it easier for participants to successfully trigger the dwell selection. Similarly, the SD_x values suggest that selection points were more spatially consistent across trials. This may be related to the reduced number of drop-offs observed in the detailed gaze analysis, indicating that participants were able to maintain a more stable fixation within the expanded target region before selection. Importantly, because the expansion factor increases with depth, the method also compensates for the higher precision demands associated with distant targets [5]. As a result, selection variability remains relatively stable across depth conditions, leading to consistently lower SD_x values and improved overall performance, which extends the literature on dwell-based selection [38,48,49,62].

Regarding **RQ2 (Does adapting the expansion size according to target depth improve user performance and experience in dwell-based gaze selection?)**, our results on Study 1 and Study 2 showed that DEPTH-BASED outperformed existing techniques in both performance and user experience. In Study 1, participants had lower movement

time, error rate, and SD_x, and higher throughput, as well as higher usability and lower cognitive load, compared to the other methods at 1.33m and 3m. These results show that a depth-based motor space expansion increases user performance and user experience. Study 2 further extends these findings to a more practical 3D environment. There, DEPTH-BASED was significantly faster than dwell baseline, while we did not observe any significant difference with other techniques. Moreover, in particular, it significantly reduced the error rate relative to the other methods. This is important because GAZE+PINCH [52, 54, 55] is often preferred for selection tasks over dwell for speed. Yet, our results show that DEPTH-BASED dwell expansion can reach a similar level of efficiency without requiring additional manual confirmation. This makes DEPTH-BASED especially relevant for scenarios in which the user's hands are occupied or where repeated manual confirmations would be undesirable.

User experience results also showed that DEPTH-BASED received 'excellent' SUS scores and lower workload ratings, whereas BUBBLEGAZE received the lowest usability scores and the highest workload. Together with the participant comments, these results suggest that techniques that dynamically change the active region around the gaze point may become difficult to select in dense 3D scenes [69, 71]. This interpretation is consistent with earlier concerns that scene changes or indirect target facilitation mechanisms may increase interaction complexity and negatively affect user experience in immersive selection tasks [71]. In contrast, DEPTH-BASED visually preserves the original target size while expanding the effective activation region of the target. This difference may explain why participants experienced it as more straightforward and less mentally demanding. Taken together, these findings show that the benefits of DEPTH-BASED is also effective in realistic 3D selection scenarios.

Overall, this paper shows that target depth affects the performance of dwell-based gaze selection and that this effect must be considered when designing target expansion methods for VR. By explicitly accounting for target depth, DEPTH-BASED provides a more reliable and consistent solution than conventional 2D expansion methods, bringing dwell-based gaze interaction closer to practical use in immersive 3D environments. These findings inform gaze-based selection design in VR, particularly for distant targets, in complex scenes, and hand-occupied interaction contexts.

9 Limitations and Future Work

Our studies were conducted in controlled VR settings with predefined target depths, interaction ranges, and scene layouts. Thus, the observed effects may differ in dynamic environments involving wider depth ranges, occlusion, or body movement. Importantly, the proposed technique is not inherently limited to static scenes: target depth and the nearest-neighbor constraint can be recomputed at runtime as the user or scene objects move. However, the studies presented here did not evaluate how such dynamic updates affect selection performance or user experience. In addition, both studies were conducted with the Meta Quest Pro, and the magnitude of these effects may vary across devices with different optics, eye-tracking quality, and focal characteristics.

The proposed depth function and its parameters were selected based on pilot testing and the depth ranges used in our studies. Although DEPTH-BASED improved performance across the evaluated conditions, the optimal scaling behavior may still depend on factors such as target density, dwell duration, and scene structure.

Future work can extend this method to investigate adaptive formulations in which the amount of expansion is jointly adjusted based on target density and estimated gaze stability. We recommend future studies on novel dwell-based interaction techniques to consider target depth as one of their design considerations. Another direction is to combine depth-aware expansion with adaptive dwell times or intent-aware methods [28, 47, 53] to further reduce unintended activations. Finally, evaluating such techniques in more ecologically valid VR applications, such as search, manipulation, or menu interaction tasks, would help establish broader design guidelines for gaze-based interaction in immersive environments.

10 Conclusion

This paper investigated how target depth affects dwell-based gaze selection with target expansion in virtual environments and evaluated DEPTH-BASED target expansion method that adapts the effective activation area according to target distance. Our findings showed that 2D expansion methods do not provide consistent benefits across different depths, highlighting depth as a critical factor in the design of gaze-based selection techniques for 3D environments. Across two user studies, DEPTH-BASED provided more robust and consistent performance than existing methods, particularly for targets positioned farther from the user. These results demonstrate the importance of incorporating depth into the design of dwell-based gaze interaction techniques for 3D environments.

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